Ministry of Communications and Information Technology

DIGITAL EGYPT CUBS

Offline Session 1:

1. Introduction To Computers and Hardware Basics
2. Inside the Computer
3. Software Basics and Operating Systems
4. Algorithms & Problem Solving

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وزارة الاتصـــــالات
وتكنولوجيا المعلومات

أشبال مصر الرقمية

مرحبًا بكم في مبادرة أشبال مصر الرقمية

المنحة المجانية المقدمة من وزارة الاتصالات وتكنولوجيا المعلومات

والتي تشارك في تنفيذها شركة iSchool الرائدة في مجال تكنولوجيا التعليم

الجلسة التطبيقية: الاولى

المستوى: الأول

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Time for an Icebreaker!

Input vs Output Device Challenge

Divide students into 2 groups:

Group 1: Input Devices Team
Group 2: Output Devices Team

Each group has 2 minutes to ask questions and collect correct devices. 1 point per answer!

Group 1 Instruction:

Ask Group 2 questions where the answer is an **input device**.

Example:
"What device do we use to type letters into the computer?"

Answer: Keyboard

Group 2 Instruction:

Ask Group 1 questions where the answer is an **output device**.

Example:
"What device shows images and text from the computer?"

Answer: Monitor

Our Session Agenda

1. **Ice Breaker: Two Truths & a Tech Lie**
2. **What is a computer?** Definition, types & evolution
3. **IPO Model** How computers process data
4. **Hardware & Input / Output** Components, input & output devices
5. **Inside the Computer** CPU, GPU, RAM, Motherboard, Storage
6. **OS & Software** Operating systems, GUI, software types
7. **Algorithms & Problem Solving** Pseudocode, flowcharts, debugging
8. **Check your knowledge**
9. **Attendance**
10. **Break Time** A 10-minute break.
11. **Team Activity** 1- Build & Run the School App
12. **Break Time** A 10-minute break.
13. **Attendance**
14. **Team Activity** 2- "The Computer Hospital" — Developer To-Do
15. **Key Takeaways** Reviewing what we've learned.
16. **Feedback & Survey**

What is a Computer?

A computer is a machine that can:

Take in information **(input)**

Work with and store information **(process & store)**

Show results **(output)**

It uses step-by-step instructions (programs) to complete tasks accurately.

Phones, tablets, and laptops all work in a similar way.

Different Types of Computers

Not all computers are the same; they come in various forms, each designed for specific purposes and portability.

Desktops

Designed for fixed locations, these powerful machines are perfect for work, study, and entertainment at home or school.

Laptops

Compact and portable, laptops offer full computing power on the go, ideal for flexible work or study from anywhere.

Smartphones & Tablets

These pocket-sized powerhouses keep us connected, browse the web, run apps, play games, and capture memories.

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The Birth of Modern Computers: ENIAC

Early electronic computers were colossal machines.

The **ENIAC** was an immense, groundbreaking pioneer:

It **filled an entire room**.

- Required large teams and vast electricity.
- Designed for mathematical calculations.

ENIAC marked the true beginning of the computer age.

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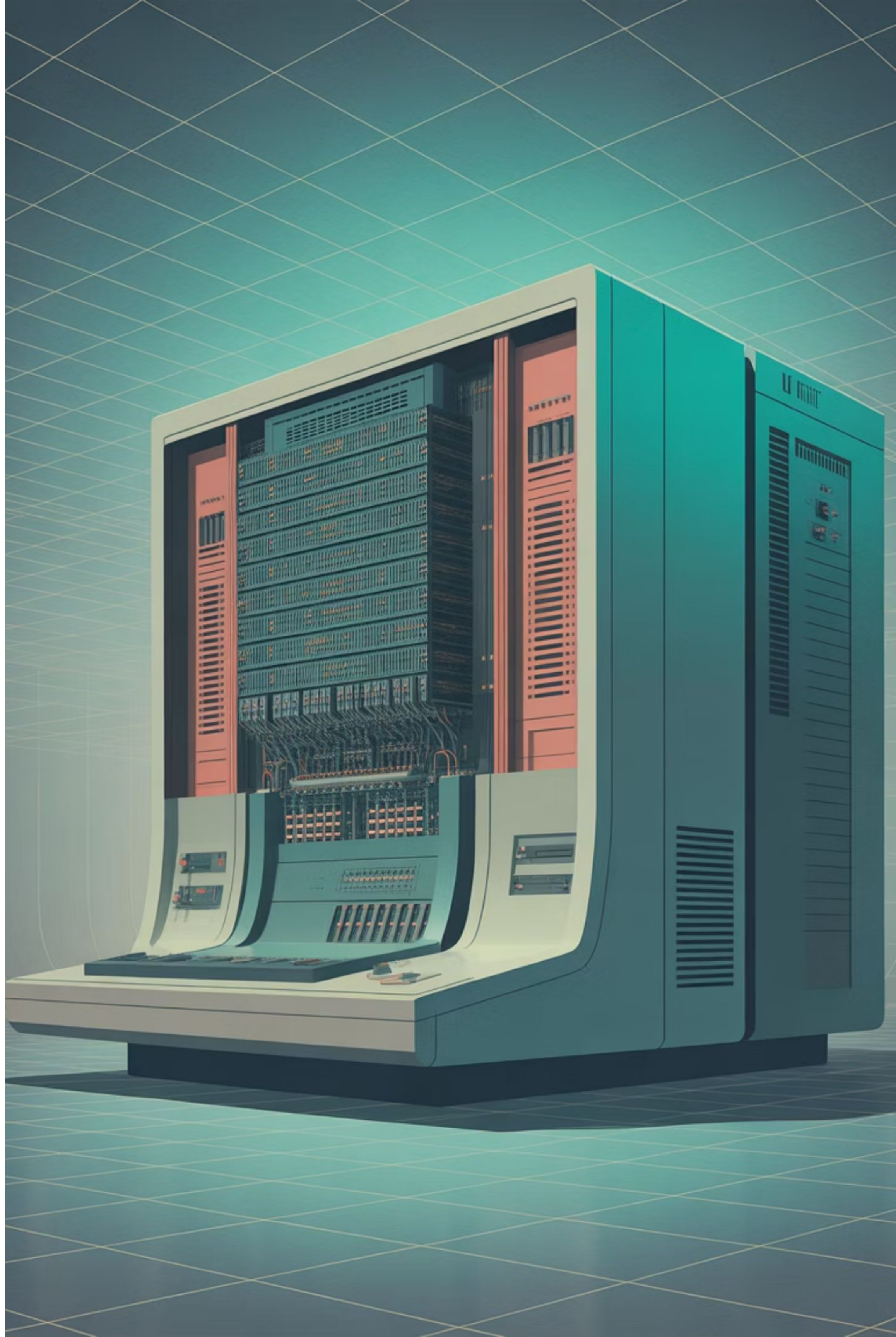
UNIVAC: Taking Computers to the World

UNIVAC (Universal Automatic Computer) emerged after ENIAC, marking a major shift in computing.

Designed for commercial and government use, it processed vast data like census and business records.

It proved computers were essential for everyday tasks, paving the way for widespread societal adoption.



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The Transistor Revolution

Early computers like ENIAC used bulky vacuum tubes that generated excessive heat and frequently failed.

The 1950s invention of the **transistor** introduced efficient semiconductor devices, making computers:

- Significantly smaller
- Faster in processing
- More affordable to produce

This "second generation" of computing increased accessibility for a wider range of applications.

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How Computers Process Data: The IPO Model

Computers follow a basic cycle known as the **IPO** (**I**nput - **P**rocessing - **O**utput - **S**torage) model to handle data:

Input

Data enters the computer through input devices

Processing

CPU performs calculations and operations on the data

Output

Results are displayed through output devices

Storage

Information is saved for future use

This cycle **repeats continuously** as you use your computer, sometimes in just fractions of a second!

What is Hardware?

When we talk about computers, we're actually referring to two essential components working together:

Hardware

The physical, tangible parts of a computer system that you can touch and see. This includes everything from the monitor and keyboard to the internal components like the processor and memory.

Software

The non-physical instructions or programs that tell the hardware what to do. Examples include operating systems, applications like word processors, web browsers, and video games.

Neither hardware nor software can function effectively without the other, forming a complete and dynamic computing system.

Internal Hardware Components

Central Processing Unit (CPU)

The "brain" of the computer that performs calculations and executes instructions.

Memory (RAM)

Temporary storage that helps the computer run programs quickly.

Storage

Long-term storage that keeps your files, apps, and system even when the computer is off.

Motherboard

The main circuit board that connects all the components together.

These components are housed inside the computer case, which protects them and keeps everything organized.

Input Devices

Input devices are hardware components that allow us to communicate with the computer by sending information into it.

They convert our actions (like typing, clicking, or speaking) into data that the computer can understand and process.

Without input devices, we wouldn't be able to tell computers what we want them to do!

Common Input Devices

Computer Scanner

Converts physical documents or photos into digital files in the computer.

Barcode Scanner

Reads barcodes to quickly input product or item data.

Touchscreen

Allows direct interaction by touching the display with your fingers.

Biometric Scanner

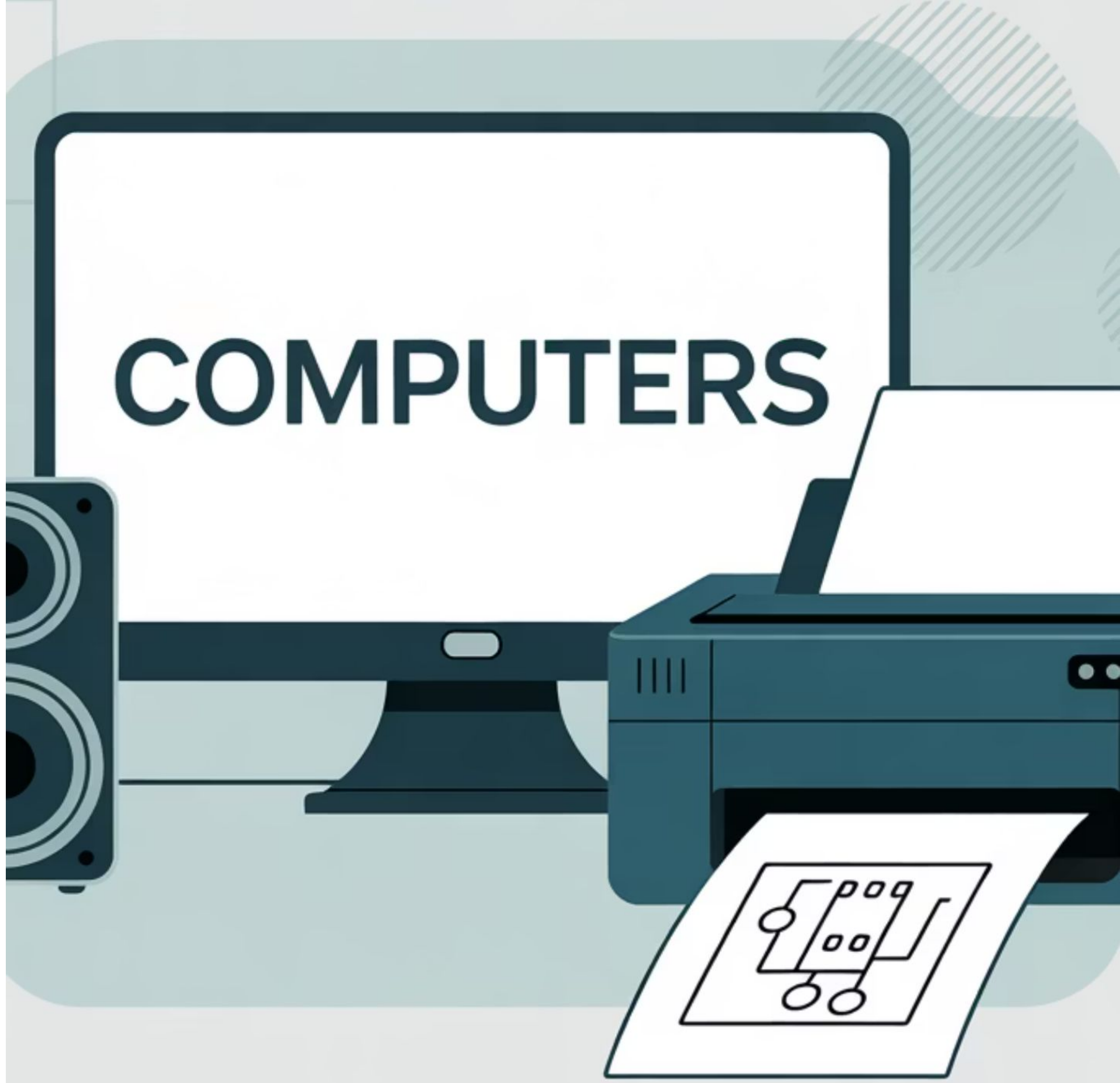
Uses unique physical features (like fingerprints or face scans) for authentication instead of passwords or PINs.

Output Devices

Output devices are hardware components that allow the computer to **communicate with us** by presenting information in a form we can understand.

They convert the computer's processed data into **text, images, sound,** or other formats that humans can perceive.

Output devices complete the communication cycle between users and computers!



Modern Output Devices

Not all output devices are traditional. They keep evolving to create immersive experiences and deliver information in new ways.

VR Headsets

Immerse you in computer-created worlds, allowing you to see and hear virtual realities as if you're inside them.

Smartwatches

Provide small screens on your wrist with apps, messages, and health information, bringing computing to your body.

Instead of only showing results on a monitor, computers can now create 3D experiences or give us information about our health and more right on our wrist.

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What is a Computer Case?

The computer case acts as the system's armor, safeguarding its sensitive internal components. It provides crucial protection from dust, physical damage, and overheating.

Additionally, the case helps manage airflow, keeping parts cool and ensuring efficient operation. Without it, components would be exposed, prone to damage, and less efficient at maintaining safe temperatures.

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Inside the Computer Case

While a computer case appears simple from the outside, it's actually designed to house and protect all the essential, complex components that make your system run. Think of it as the foundational structure that keeps everything vital safe and organized.

Motherboard	**CPU**
RAM	**Storage Drives**
Power Supply	**GPU**

The case ensures these critical parts are organized, protected, and properly connected to work together seamlessly.

Desktop vs Laptop Cases

Desktop Cases

Desktop cases are large towers, offering ample space for powerful components and extensive upgrades. They connect to external monitors, keyboards, and mice, ideal for customization and high performance.

Laptop Cases

Laptops integrate all components, including screen and keyboard, into a single, portable unit. While convenient for mobility, their compact design limits significant upgrades, making them perfect for computing on the go.

Evolution of Case Designs

If you look at older desktop cases, you'll notice a slot in the front for CDs and DVDs. Today, most cases no longer have them. This transition highlights how case designs evolve with technology, adapting to new storage methods and changing user needs over time.

CD/DVD

CD and DVD drives were once essential components of every computer. People used them to:

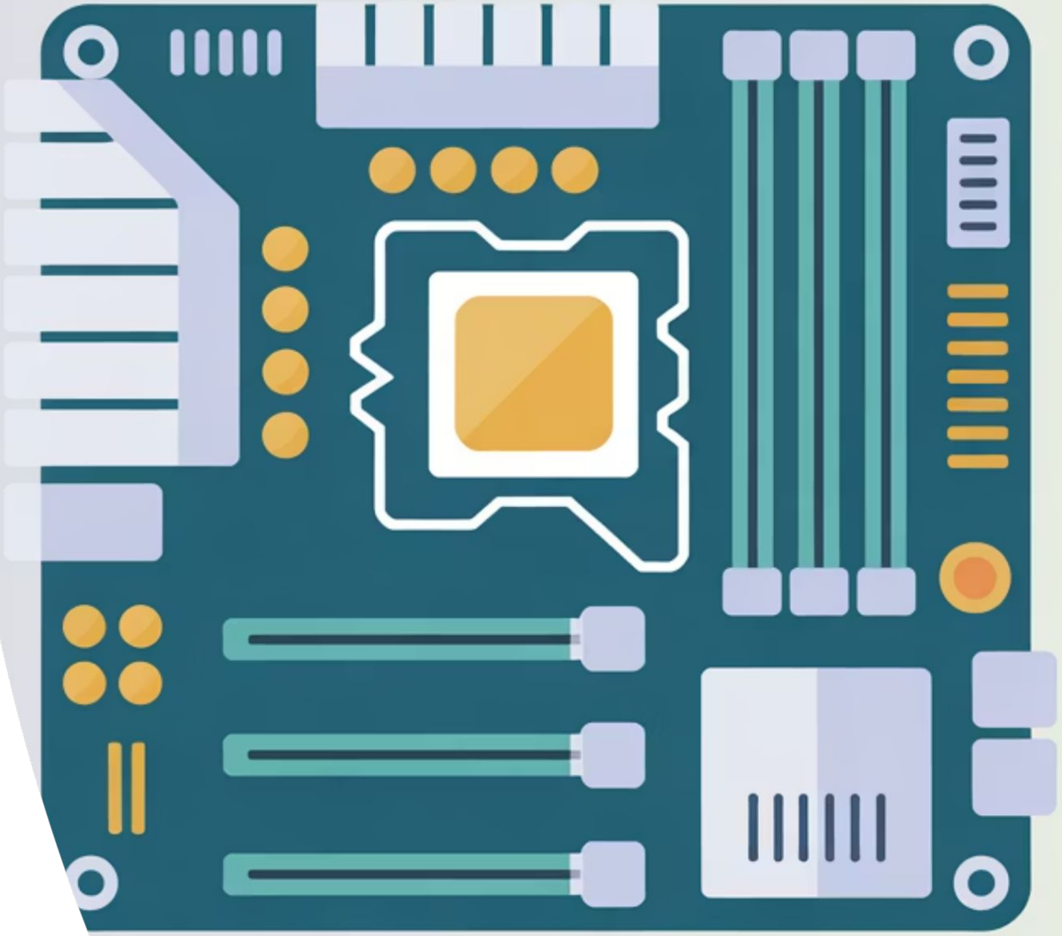
- Install software and operating systems
- Play music CDs and watch DVD movies
- Backup important files to physical discs

That's why older desktop cases almost always had a slot for them.

What is a Motherboard?

The **motherboard** is the computer's central system, acting as the "main board" that holds everything together. It's a large circuit board that connects crucial components like the Central Processing Unit (CPU), memory, and storage drives.

Think of it as the foundation and primary communication hub, directing power and data flow to ensure all parts function and work in harmony.



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Circuits

Imagine the motherboard like a bustling city map, where the intricate lines are not just roads, but vital **circuits** carrying data from one point to another. When the **CPU** needs information from **RAM**, for example, the signals efficiently travel along these precisely laid **pathways**.

This makes the motherboard the computer's central **"traffic system"**, ensuring seamless communication and operation between all connected components.

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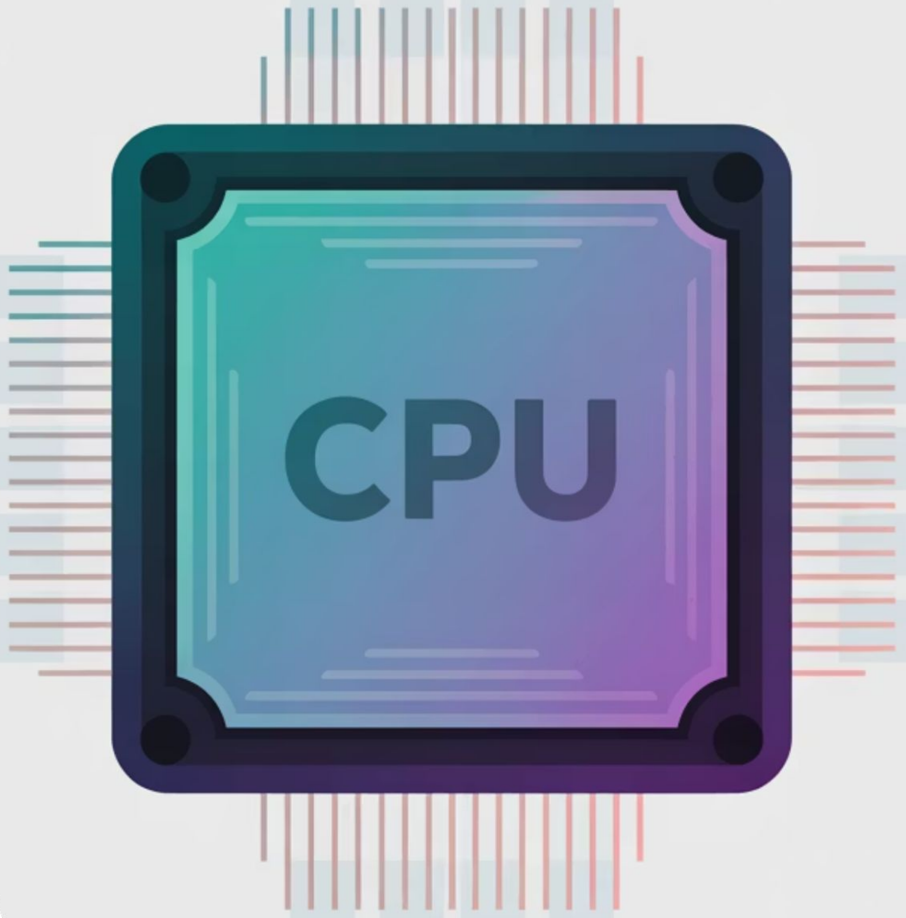
What is a CPU?

The **Central Processing Unit (CPU)** acts as the **computer's brain.** It is primarily responsible for executing instructions and performing calculations. The CPU processes all information and runs every program, making it essential for nearly all computer operations.



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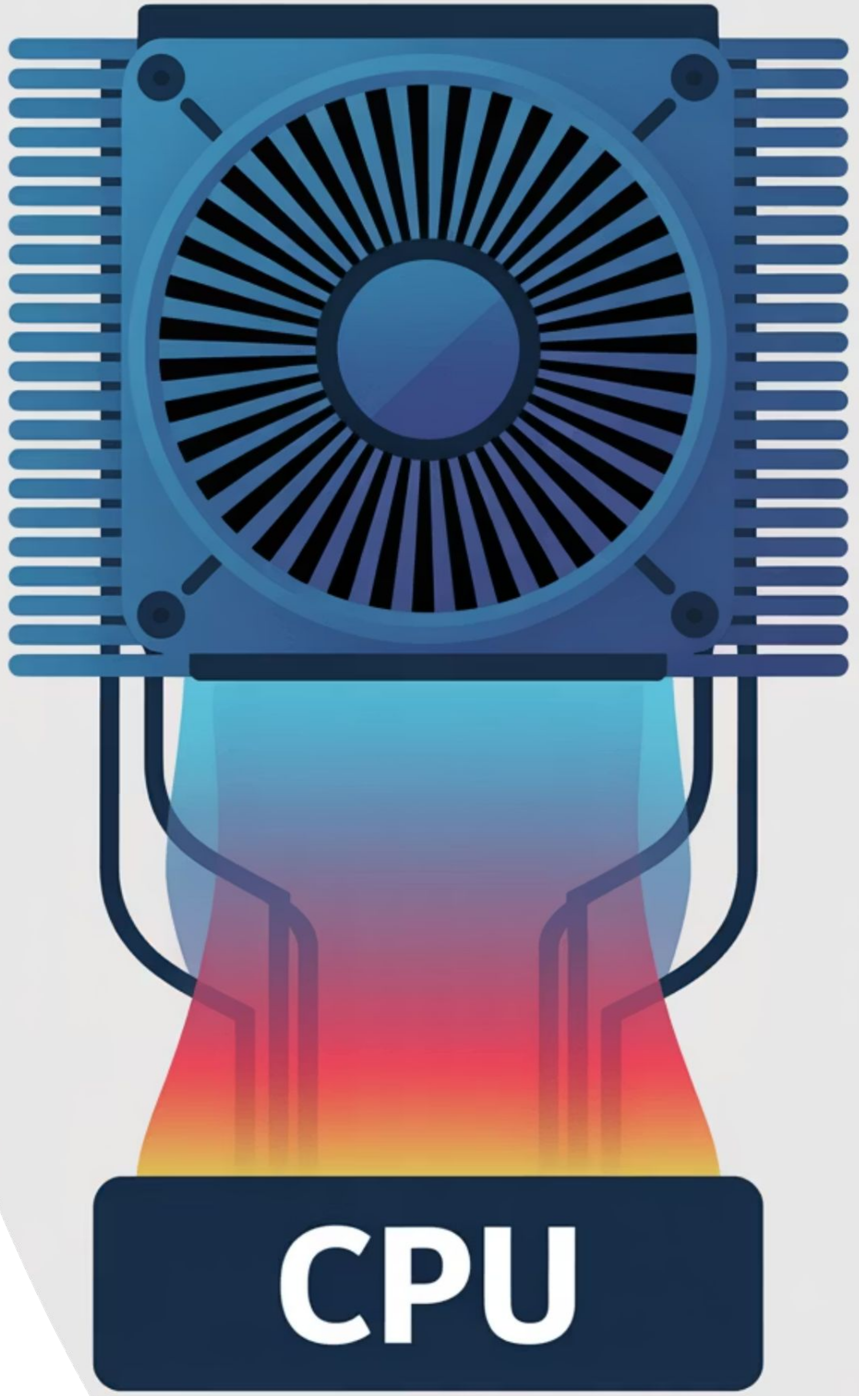
What is a Heatsink?

A **heatsink** prevents electronic devices, like the CPU, from overheating. It transfers heat away, often with a fan, to keep crucial computer components cool and operating effectively.



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What is a GPU?

The **GPU** specializes in rendering pictures, animations, and complex 3D worlds. While the CPU handles general tasks, the GPU is optimized for graphics.

Essential for video games and professional editing, it ensures smoother graphics and faster rendering for high visual fidelity.

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The Power of Synergy: CPU, GPU, and Motherboard

All three essential parts work as a cohesive team. The **motherboard** serves as the highway system, facilitating data flow, while the **CPU** acts as the general brain, handling core computations, and the **GPU** is the specialized graphics brain.

If any one of these components is a bottleneck, the entire computer's performance can significantly slow down. This is why users, especially gamers and creative professionals, often prioritize investing in powerful CPUs and GPUs to ensure optimal performance for demanding tasks.

CPU & GPU: The Price-Performance Balance

Basic Laptop / Everyday Use

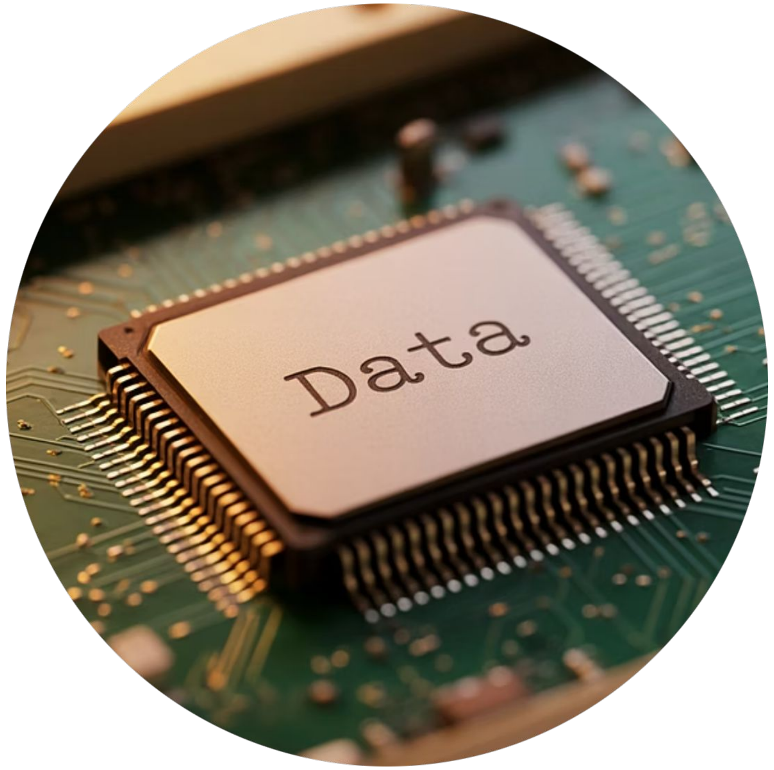
For general tasks, these use a **simple CPU** and integrated graphics. They are **affordable** but offer **slower performance** for demanding applications.

Gaming PC / Workstation

For complex tasks like gaming or video editing, these have a **powerful CPU** and **dedicated GPU**, offering **faster performance** at a **higher price tag**.

Your choice of computer fundamentally depends on your specific needs and intended use, directly influencing the required CPU and GPU power.

Types of Computer Memory



RAM (Random Access Memory)

RAM is where your computer stores data it's actively using. It's temporary memory; once you turn off the computer, everything in RAM is erased. Think of it like a desk: the bigger the desk, the more tasks you can manage simultaneously. If RAM gets full, your computer slows down, much like a cluttered desk.

ROM (Read-Only Memory)

ROM is a non-volatile memory that stores essential startup instructions, like a computer's "recipe book." Unlike RAM, its contents are permanent and don't erase when the power is off. It's crucial for the computer to know how to boot up and load the operating system.

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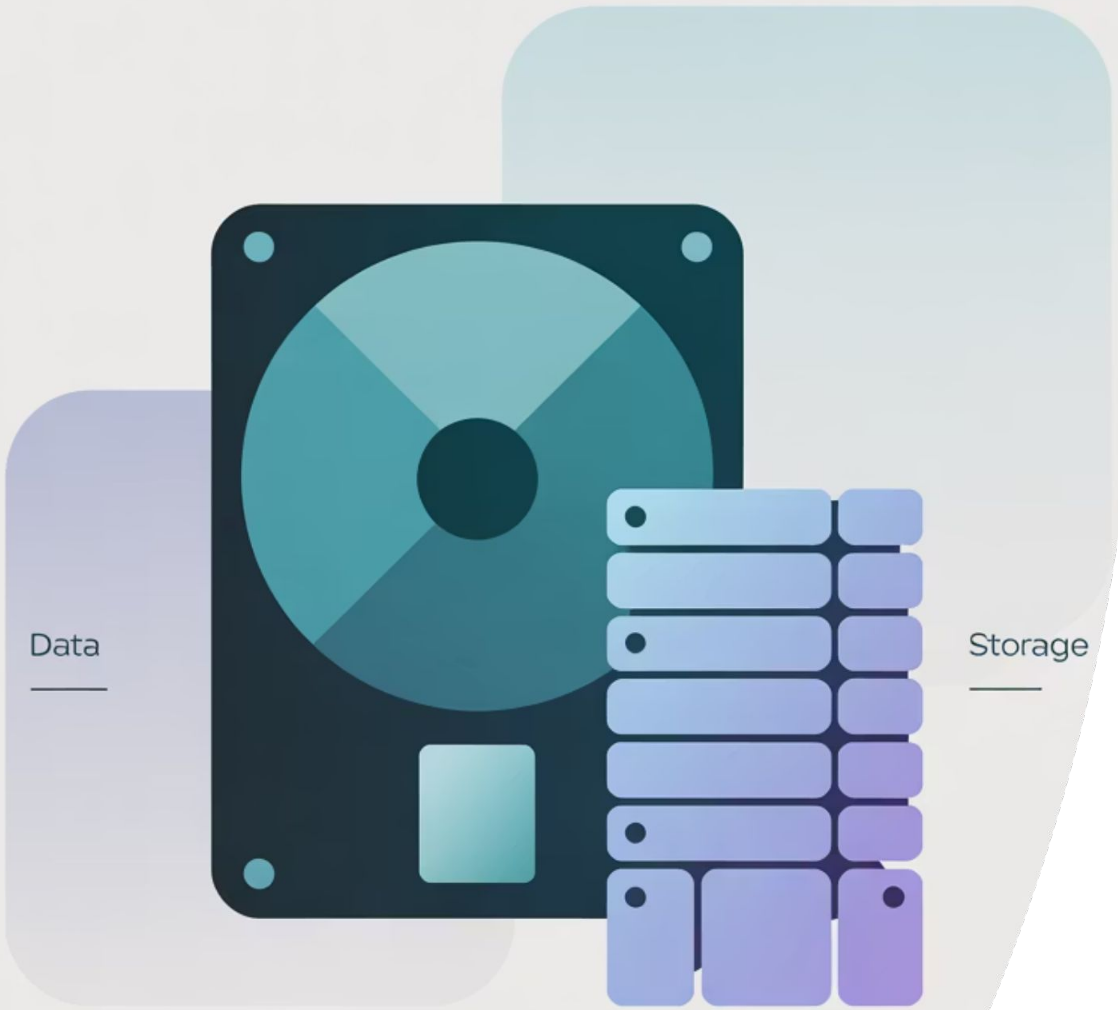

What is Storage?

Storage serves as the computer's long-term memory, holding data persistently even when the power is turned off. Unlike the temporary nature of RAM, storage is where your operating system, applications, music, photos, and documents are permanently kept.

Today, the most common forms of computer storage are traditional **Hard Disk Drives (HDDs)** and the faster, more modern **Solid-State Drives (SSDs)**.

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Types of Storage

Hard Disk Drive (HDD)

HDDs are traditional storage devices that use spinning platters to store data. They offer large storage capacities at a lower cost, making them suitable for archiving and bulk data.

Solid State Drive (SSD)

SSDs are faster, more durable storage devices that use flash memory. They have no moving parts, resulting in quicker boot times and application loading, ideal for operating systems and frequently used programs.

External Storage

This refers to devices like external hard drives, USB flash drives, or cloud storage, used for portability, backups, or expanding a computer's storage capacity.

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Memory vs. Storage: A Key Distinction

Memory (RAM)

RAM is your computer's **short-term memory**, holding active data for quick access and multitasking, directly impacting speed.

Storage (HDD/SSD)

Storage is your computer's **long-term memory**, permanently saving operating system, applications, and files, even when powered off.

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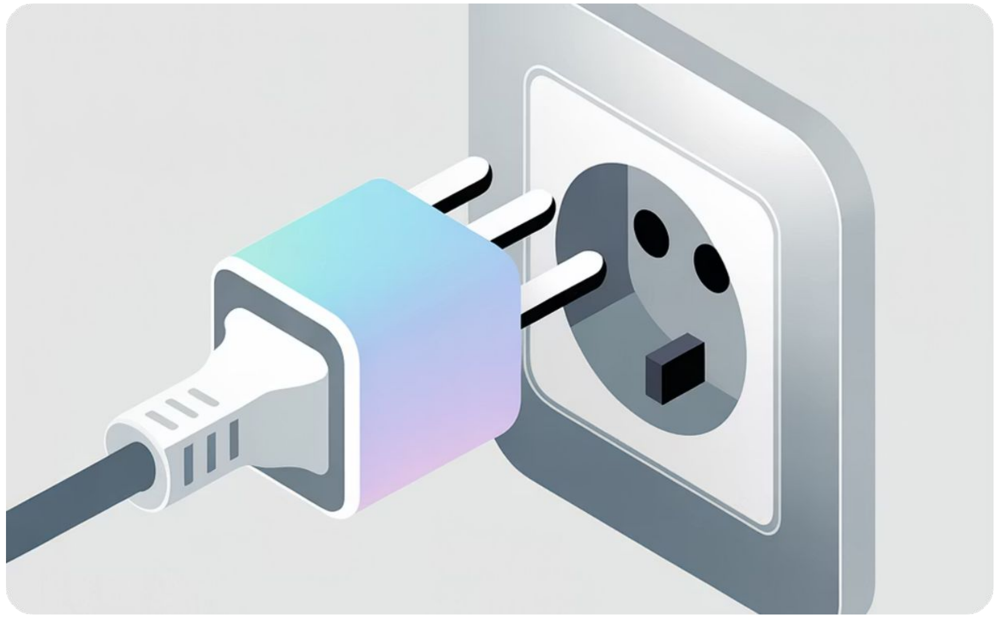
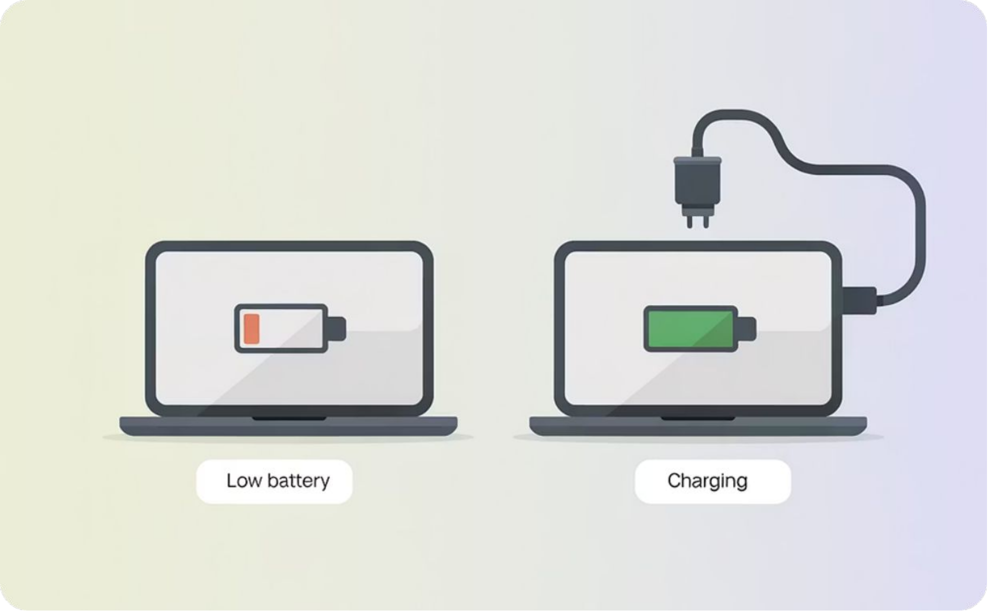
Power Supply & Batteries

The **power supply** changes electricity from your wall outlet into power your computer can use.

Computers get their power in two main ways:





Plugged-in Power

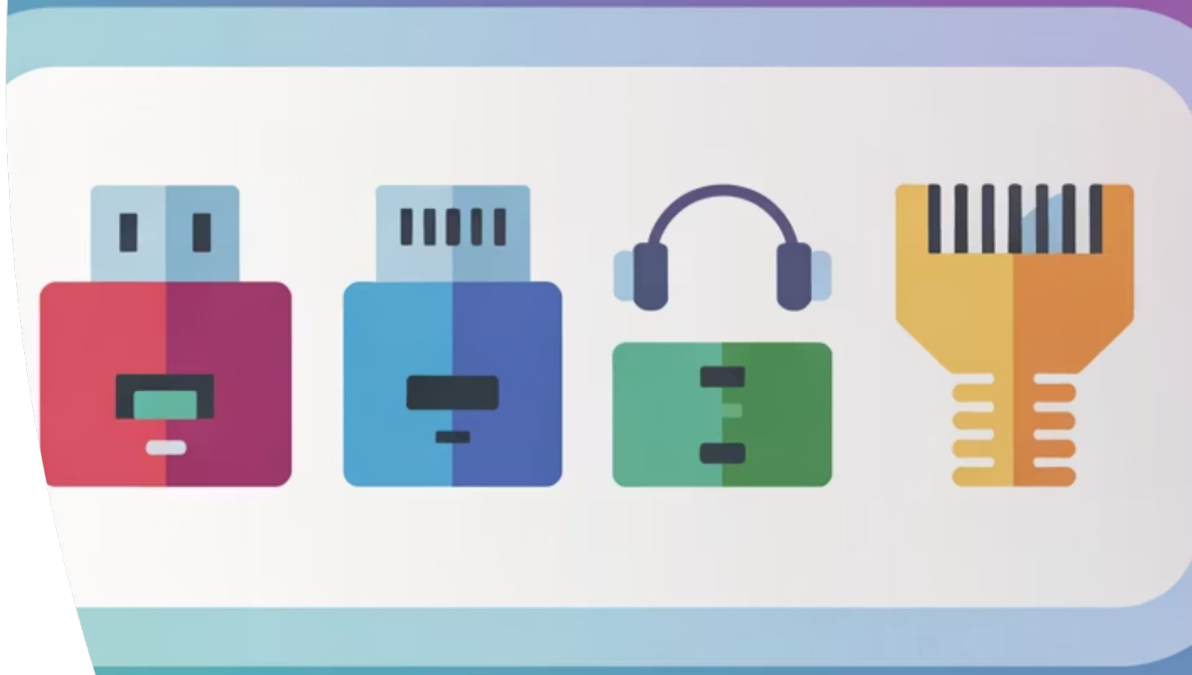
This uses power from a wall outlet. It gives a steady and endless supply, great for desktops and charging laptops.

Batteries

Batteries are for devices you can carry, like laptops and phones. They let you use your device without being plugged in, but only for a limited time.

Ports: Your Computer's Connections

Ports are like the doors and windows of your computer, allowing it to connect with the outside world and other devices. They are crucial physical interfaces that enable your computer to communicate with and extend its capabilities by connecting peripherals, external storage, displays, networks, and more.



Ports

USB Port

Used to connect many different devices like keyboards, mice, printers, and flash drives.

HDMI Port

Connects your computer to external displays like monitors or TVs for high-definition video and audio.

Headphone Port

Allows you to plug in headphones or speakers to listen to audio from your computer.

Ethernet Port

Connects your computer to a wired network or the internet for a fast and stable connection.

What is Software?

Software is a **set of instructions that tells a computer** what to do It's **the "brain" that makes hardware useful.**

Without software, computers would just be expensive paperweights.

Software is what brings your computer to life, allowing you to play games, write documents, and browse the internet.

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Types of Software

System Software

Essential programs manage a computer's hardware and software, ensuring everything runs smoothly. It includes Operating Systems, device drivers, and Utilities.

Application Software

These programs are used for specific tasks such as browsing the internet, playing games, or creating documents.

Programming Software

These tools are utilized by developers to create, compile, and debug.

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The Dawn of Software: Binary Code

The first major leap towards modern software came with

binary code.

Computers only understand: on (1) and off (0). Translating instructions into these sequences allowed humans to **program machines without physical rewiring**.

Binary was **complex** and **time-consuming** to write.

The Dawn of Data Entry: Punched Cards

Punched cards were stiff paper used to store data. Computers read them by checking for holes, allowing instructions without typing 0s and 1s.

Originating in 1800s weaving and the 1890 U.S. Census, they were vital for entering data and running programs for years, leaving a major mark on computing history.

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The Next Step: Assembly Language

After binary and punched cards, low-level programming languages appeared. **Assembly language** let programmers use short codes or symbols instead of long rows of 1s and 0s. This made software easier to write, but it still required deep technical knowledge.

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High-Level Languages

The big breakthrough came with **high-level programming languages**.

Instead of using only numbers or short codes, programmers could now write using familiar words and phrases, much like human language. Keywords like `print`, `if`, and `loop` made code far more intuitive.

Languages such as **BASIC**, **C**, **Java**, and **Python** are prime examples. They made programming significantly **faster** and **easier to learn**, truly opening the door to the modern software world.

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What is the Operating System (OS)?

The OS is the **most important system software** on your computer

- It manages all hardware resources and provides services for other programs

Popular examples include **Windows**, **macOS**, **Linux**, and **Android**

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Versions

Operating systems are regularly updated, and each new version brings important improvements in security, performance, and new features.

These continuous updates are crucial for keeping your devices safe, stable, and equipped with the latest technology.

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Resource Management by OS

The operating system is like a traffic controller for your computer's resources. It makes sure that:

- Multiple programs can run at the same time without crashing
- Each program gets its fair share of processor time
- Programs don't interfere with each other's memory
- Files are stored and retrieved correctly

Without proper resource management, your computer would become slow, crash frequently, or even stop working altogether!

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Drivers

The OS talks to devices using drivers — small pieces of software that help devices understand commands. It's the language that the OS can communicate with for each device connected to it.

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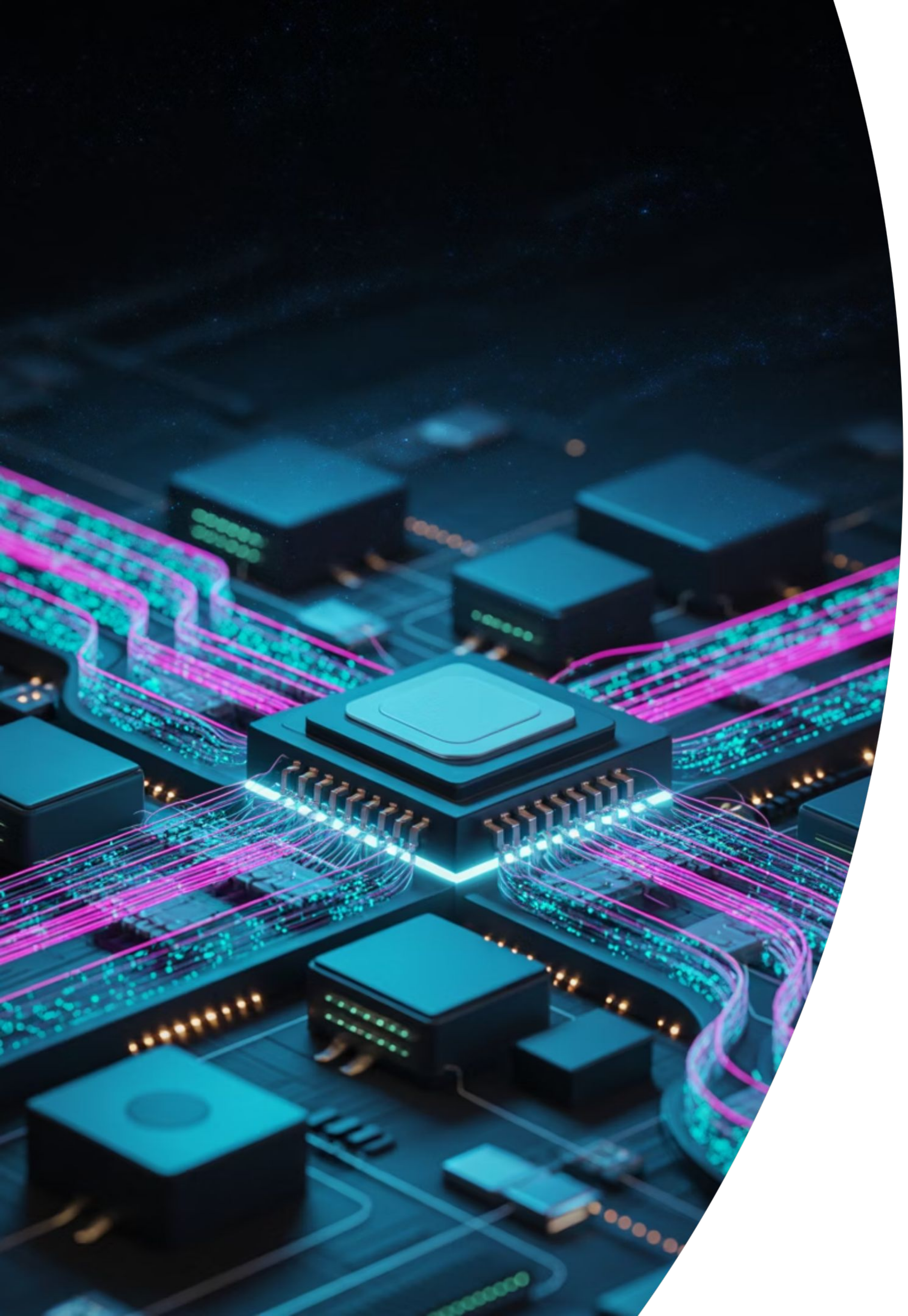


Data Buses: The Information Highways

Data buses are vital digital pathways connecting computer components like the CPU and RAM, and external devices. The operating system (OS) manages this data flow, acting as a traffic controller to ensure efficient transfer and prevent performance-slowing congestion.

Different Bus Types: Specialized Pathways

Different data buses connect specific components for efficient data flow.

Memory Bus

Connects CPU to RAM for rapid data exchange, crucial for program execution.

Storage Bus

Links CPU to storage (HDDs, SSDs) for managing file saving and retrieval.

Expansion Bus

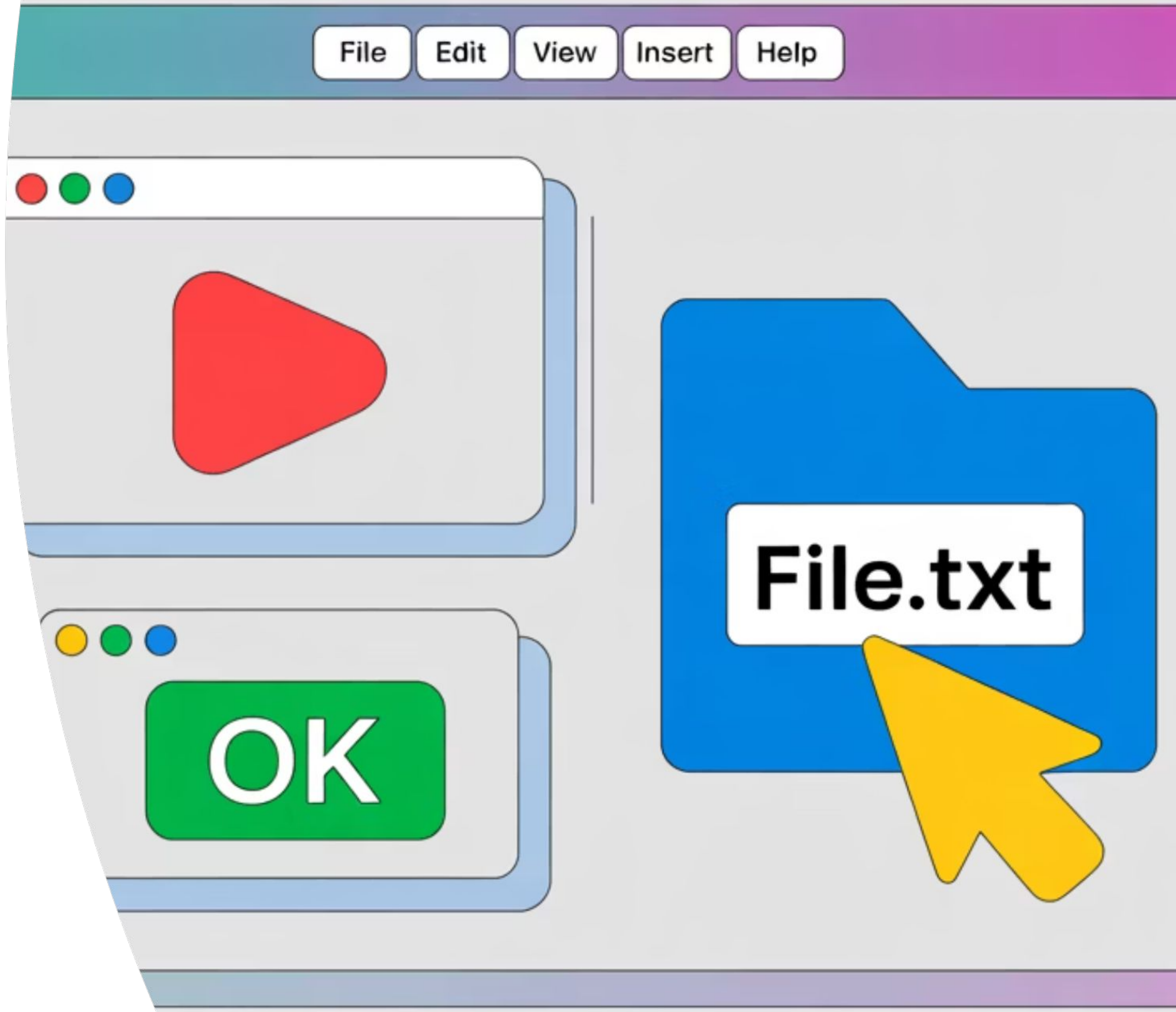
Connects CPU to peripherals like printers, keyboards, and external devices.

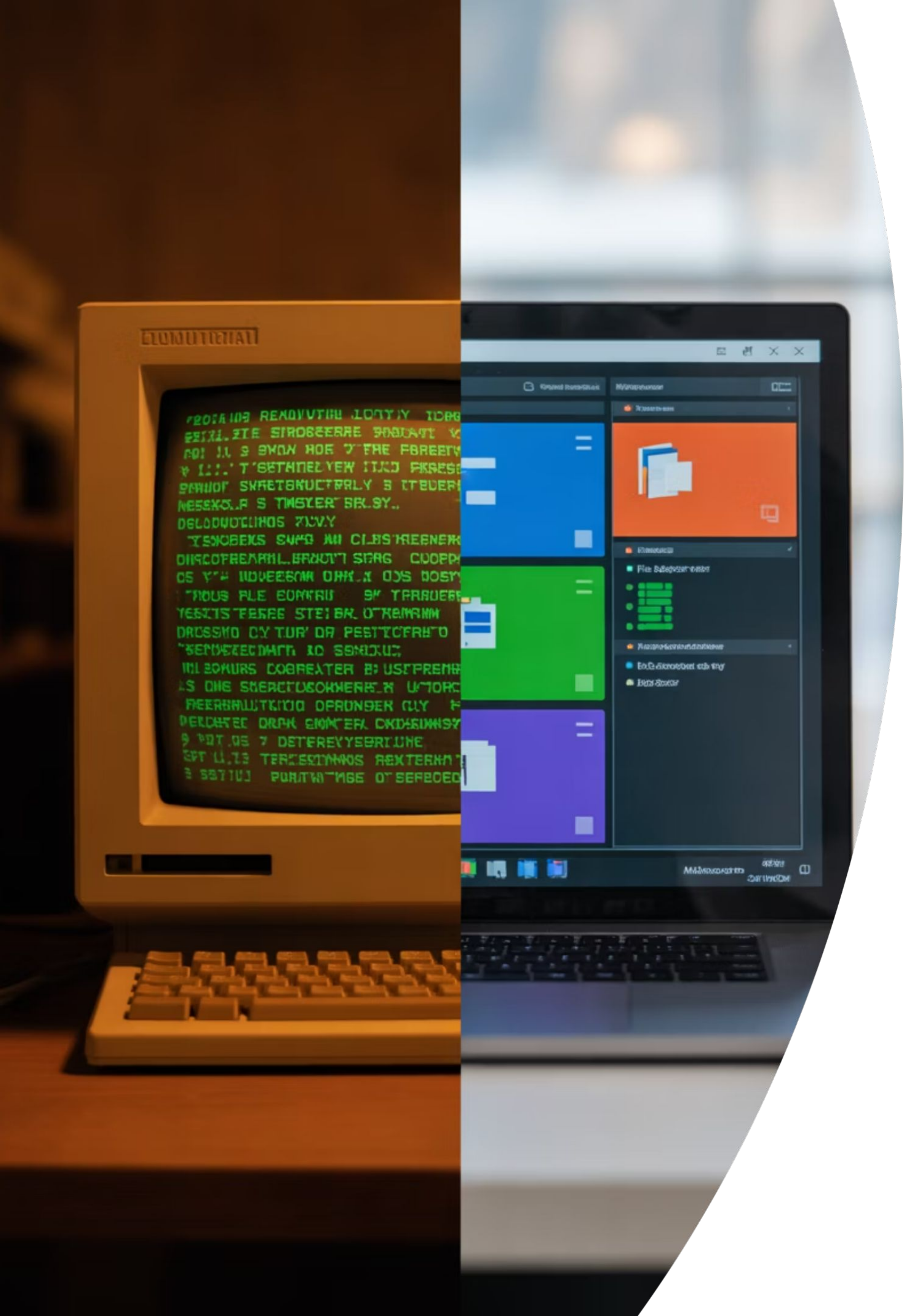
Think of it like a city with specialized roads for memory, storage, and peripherals. The OS acts as the central traffic controller, ensuring data reaches its correct destination.

Graphical User Interface (GUI)

The GUI is how you interact with your computer using visual elements instead of typing commands:

- Icons represent programs and files
- Windows allow you to view content and run programs
- Menus provide options organized in categories
- Buttons perform actions when clicked
- Pointers (mouse cursors) help you select and interact





A Revolution in Accessibility

Graphical User Interfaces (GUIs) revolutionized computing by making it accessible to everyone. They replaced complex text commands with intuitive visual elements, transforming a technical, exclusive domain into an approachable digital world for a global audience.

GUI Elements Up Close

Icons

Small pictures that represent programs, files, or functions. Clicking them usually opens or activates what they represent.

Windows

Rectangular areas on the screen that display content or run programs. You can often move, resize, or close them.

Menus

Lists of options or commands, usually found at the top of a window or screen. They help organize actions you can perform.

What Is an Algorithm?

An **algorithm** is a set of clear, step-by-step instructions used to solve a problem or complete a task — used in daily life *and* by computers.

Clarity

Must be clear enough for anyone to follow and get the same result

Sequencing

Steps must be in the correct order — mixing them up leads to wrong results

Precision

Computers follow steps exactly — mistakes in sequence can't be self-corrected

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What Is Pseudocode?

Writing logic in plain human language

It's your **rough draft** before writing real code.

Use Plain English

Write steps as if explaining to a friend.

- Anyone can read it — no coding knowledge needed

Show the Logic

Use simple logical markers.

- IF, THEN, ELSE for decisions
- REPEAT/WHILE for loops

No Syntax Rules

Focus on logic, not programming language rules.

- No semicolons or brackets required
- No strict grammar rules

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What Is a Flowchart?

A **flowchart** is a diagram that shows the steps of a process using shapes connected by arrows. Learn the shapes once — read any algorithm diagram instantly.

Oval
Start / End

Rectangle
Process / Action

Diamond
Decision (Yes/No)

Parallelogram
Input / Output

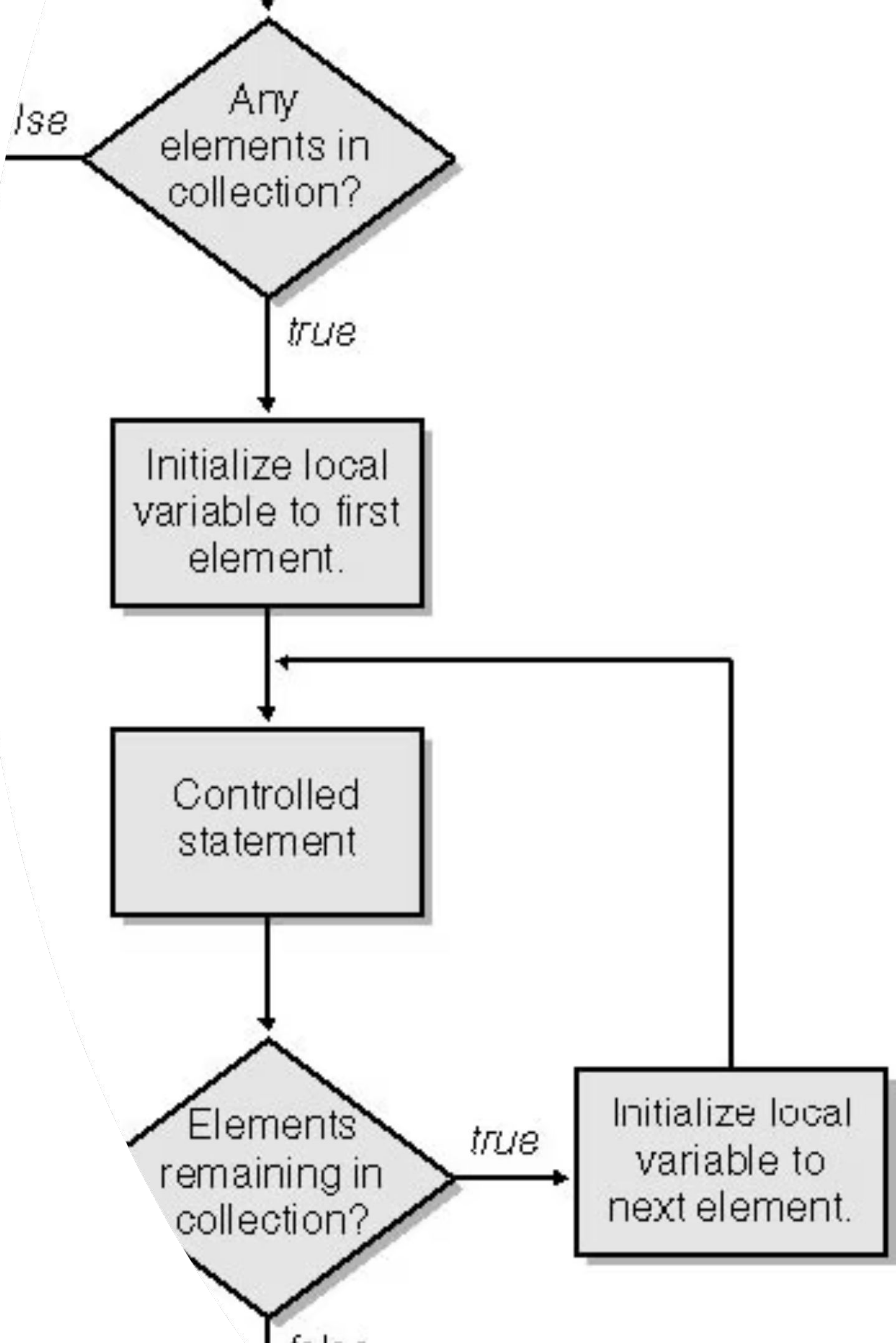
Arrow
Direction of flow

Loops in Flowcharts

A loop allows a set of instructions to be repeated multiple times, either for a specific count or until a certain condition is met. It's essential for tasks that require iterative actions.

The diamond shape (decision) is crucial for loops. It checks if a condition is true or false, directing the flow back to an earlier step (the loop) or forward to the next part of the process.

This example shows the loop-back path in action: if the answer is true, the flow returns to process the next item; if false, the process moves on to End.



Problem-Solving Framework

Understand

Know exactly what you're trying to solve before starting

Break It Down

Split big problems into smaller, simpler steps

Plan Your Steps

Use pseudocode or flowcharts to plan carefully

Be Specific

Each step must be clear and precise — avoid vague instructions

Test & Improve

Run your solution with different inputs and fix issues

Debugging Algorithms

Debugging = finding and fixing errors (bugs) in an algorithm. Even the best engineers debug daily — it's part of the process, not a failure.

🚫 Bug Type 1

Missing START or END — algorithm has no clear boundaries

🔀 Bug Type 2

Decision diamond missing one path — every diamond MUST have both YES and NO exits

❓ Bug Type 3

Using a variable before it's defined — can't use `age` if you never asked for it

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Questions & Discussion 💬

Got a question? We'd love to hear it!

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Let's Check your knowledge

🧠 Quick Check: Multiple Choice Challenge

Time to test your knowledge! Choose the best answer for each question based on what we've covered today.

1

What does IPO stand for, and which component handles the 'Processing' step?

(A) Input, Print, Output — handled by RAM

(B) Input, Process, Output — handled by CPU

(C) Interface, Process, Output — handled by GPU

(D) Input, Process, Output — handled by SSD

2

A student says: 'RAM stores my files even after I shut down.' What is wrong?

(A) RAM is permanent; storage is temporary

(B) Nothing — the student is correct

(C) Both RAM and storage lose data on shutdown

(D) RAM is temporary — data is lost on shutdown

3

What is the PRIMARY role of an Operating System?

(A) Manage hardware resources & run programs

(B) Store user documents permanently

(C) Render 3D game graphics

(D) Connect to the internet

🧠 Quick Check: Multiple Choice Challenge

Time to test your knowledge! Choose the best answer for each question based on what we've covered today.

1

Why is debugging important?

(A) It deletes all files from the computer.

(B) It replaces the operating system.

(C) It helps find and fix errors in an algorithm or program.

(D) It turns hardware into software.

2

Which flowchart shape is usually used for a decision?

(A) Diamond

(B) Oval

(C) Rectangle

(D) Arrow

3

What is software?

(A) A physical part that can be touched.

(B) A type of storage cable.

(C) A set of instructions that tells the computer what to do.

(D) A device used only for sound.

Attendance

فترة الاستراحة

🛠️ Team Activity 1: Build & Run the School App

👥 Teams of 3–4	⏱️ 20 Minutes

Scenario: A school has hired your team to design and set up a digital attendance system. Your team must choose the right hardware for it, explain how the OS will manage it, and write the algorithm that runs it — then present the full plan to the class.

Task 1 — Hardware Station

Open your team's FigJam board. In the Hardware Station frame, you'll find 12 component cards: CPU, RAM, GPU, SSD, HDD, Motherboard, PSU, Monitor, Keyboard, Webcam, Printer, USB Port.

Drag each card into one of two columns:

- ✅ Need It — components the school's attendance system actually requires
- ❌ Don't Need It — components that are unnecessary for this task

For every card in the ✅ Need It column, add one sentence explaining why it is needed.

🛠️ Team Activity 1: Build & Run the School App

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Scenario: A school has hired your team to design and set up a digital attendance system. Your team must choose the right hardware for it, explain how the OS will manage it, and write the algorithm that runs it — then present the full plan to the class.

Task 2 — OS Station

In the OS Station frame, read this scenario pinned at the top:

“The app is running a webcam to record attendance + saving data to SSD + printing a report — all at the same time.”

Fill in the pre-built table on the board with 3 rows — one per process

Then answer the bonus question in the text box below the table: “If RAM is only 2GB — what happens and what does the OS do?”

🛠️ Team Activity 1: Build & Run the School App

👥 Teams of 3–4	⏱️ 20 Minutes

Scenario:A school has hired your team to design and set up a digital attendance system. Your team must choose the right hardware for it, explain how the OS will manage it, and write the algorithm that runs it — then present the full plan to the class.

Task 3 — Algorithm Station

In the Algorithm Station frame, write the pseudocode for recording one student's attendance. Your pseudocode must include(**INPUT** — collect the student's ID, **IF / ELSE** — check if the student is on the registered list, **OUTPUT / DISPLAY** — show "Present" or "Not Registered") Then draw the flowchart for your pseudocode in the blank frame next to it.

🛠️ Team Activity 1: Build & Run the School App

👥 Teams of 3–4	⏱️ 20 Minutes

Scenario: A school has hired your team to design and set up a digital attendance system. Your team must choose the right hardware for it, explain how the OS will manage it, and write the algorithm that runs it — then present the full plan to the class.

Task 4: Team Presentation

Screen-share your completed FigJam board. Each team member presents one station — explains their choices and reads their justifications. Other teams give ✅ if they agree with a decision or 🔄 if they would change something, with a reason.

📝**Team Roles:** 🖥️ Hardware Lead — sorts the component cards and writes the justification sticky notes in Task 1, ⚙️ OS Manager — fills in the OS table and answers the bonus question in Task 2,
✏️ Algorithm Writer — writes the pseudocode and draws the flowchart in Task 3, 🎤 Spokesperson — leads the presentation and coordinates between all three stations
🔗**Link:** https://www.figma.com/board/j29C0vFNVtl0TNQu5d4Dpv/Build---Run-the-School-App?node-id=0-1&t=0VvgCFuYD2Wc0OXz-1
📌**Important note:** Ask each group to make a copy of everything in the link, then open FigJam, paste them, and work on the activity there.

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Attendance

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🛠️ Team Activity 2: "The Computer Hospital" — Developer To-Do

👥 Teams of 3–4	⏱️ 20 Minutes

Scenario: You are the technical support team at the Computer Hospital. Four patients have just walked in — each describing a different problem with their computer or program. Your team must diagnose every case, identify which session the concept comes from, and write the prescription that fixes it.

Task 1 — Read the Patient Files

Open your team's FigJam board. In the Patient Files column, read all four patient sticky notes carefully before writing anything. As a team, discuss each case and agree on what the problem is before moving to Task 2.

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🛠️ Team Activity 2: "The Computer Hospital" — Developer To-Do

👥 Teams of 3–4	⏱️ 20 Minutes

Scenario: You are the technical support team at the Computer Hospital. Four patients have just walked in — each describing a different problem with their computer or program. Your team must diagnose every case, identify which session the concept comes from, and write the prescription that fixes it.

Task 2 — Diagnose Each Patient

In the Diagnosis column, post one sticky note per patient answering these three questions:

- What is the exact problem?
- Which session does this concept come from? (Session 1 / 2 / 3 / 4)
- Which specific component or concept is the root cause?

🛠️ Team Activity 2: "The Computer Hospital" — Developer To-Do

👥 Teams of 3–4	⏱️ 20 Minutes

Scenario: You are the technical support team at the Computer Hospital. Four patients have just walked in — each describing a different problem with their computer or program. Your team must diagnose every case, identify which session the concept comes from, and write the prescription that fixes it.

Task 3 — Write the Prescription

In the Prescription column, post one sticky note per patient with exactly two sentences:

- Sentence 1: The fix — what should be done or changed?
- Sentence 2: The prevention — how do you stop this from happening again?

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🛠️ Team Activity 2: "The Computer Hospital" — Developer To-Do

👥 Teams of 3–4	⏱️ 20 Minutes

🗒️ **Scenario:**You are the technical support team at the Computer Hospital. Four patients have just walked in — each describing a different problem with their computer or program. Your team must diagnose every case, identify which session the concept comes from, and write the prescription that fixes it.

Task 4: Team Presentation

Screen-share your completed FigJam board. The Spokesperson picks one patient — reads the diagnosis and prescription out loud. Other teams give ✅ if they agree with the diagnosis or 🔄 if they would diagnose it differently, with a reason.

📝**Team Roles:** 🦷 Doctor 1 — diagnoses Patient 1 and Patient 2 and posts their sticky notes, 🦷 Doctor 2 — diagnoses Patient 3 and Patient 4 and posts their sticky notes, 💊 Pharmacist — writes the two-sentence prescription for all four patients, 🎤 Spokesperson — presents the chosen patient’s diagnosis and prescription to the class

📌**Important note:** Ask each group to make a copy of everything in the link, then open FigJam, paste them, and work on the activity there.

🔗**Link**: https://www.figma.com/board/CdSQgB0FeoVwTPCTDmzuM5/%22The-Computer-Hospital%22-%E2%80%94-Developer-To-Do?node-id=0-1&t=iDLCumMPCewuOXTb-1

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Key Takeaways

Computers follow the IPO model for data:
Input, Processing, Output, and Storage.

Core hardware
includes the CPU, GPU, RAM, and the connecting motherboard.

RAM is temporary memory for active data;
Storage keeps files permanently.

The operating system manages:
hardware, resources, and communicates via drivers.

Algorithms are step-by-step instructions.
planned using pseudocode or flowcharts.

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Coming up next session!

- **Introduction to Programming and Computational Thinking**:
  Learn the basics of programming and how to solve problems using logical step-by-step thinking.
- **Block-Based Coding:**
  Start coding using visual blocks to understand programming concepts in an easy and interactive way.
- **Text-Based Coding**:
  Move to writing real code using text-based programming and learn the differences from block-based coding.

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1

Rate the Session

How would you rate today's session overall?

2

What Worked Best?

Which topic or activity was your favorite today?

3

What Can Improve?

Is there anything you'd like explained differently next time?

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